C++

PENG

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1 Introduction

Definition 1.1 (C++)**.**

- *C is based on functions.*
- *C++ is object-oriented: separation of abstract mathematical concepts and black-box implementation [1, 2].*

2 Types and Variables

Definition 2.1 (arithmetic type)**.**

```
char              // '\0': null, '\t': horizontal tab, '\n': newline
#include <string> // using std::string;
std::string
int               // unsigned int >= 0
bool              // true 1 and false 0
double
```

Definition 2.2 (bit)**.** *1 byte = 8 bits*

bit	*combination*	*total*	`unsigned`	`int`
1	$0, 1$	2^1	$0, 1$	$0, 1$
2	$00, 01, 10, 11$	2^2	$0, 1, 2, 3$	$0, 1, -2, -1$
...				
32		2^{32}	$0, 1, ..., 2^{32} - 1$	$0, ..., 2^{16} - 1, -2^{16}, ..., -1$

Definition 2.3 (type conversion)**.**

```
// int => unsigned
unsigned i = -1;              // -1 = 2^{32} - 1 for 32-bit unsigned int
#include <cmath>
unsigned i = std::pow(2, 32); // 2^{32} = 0 for 32-bit unsigned int
unsigned a = 1; int b = -1;
a + b // 1 + (2^{32} - 1) = 2^{32} = 0
a < b // 1 < (2^{32} - 1) true
```

2.1 Variable Initialization

Definition 2.4 (object)**.** *An object is a memory region that contains data with type.*

Definition 2.5 (initialization and assignment)**.**

- *initialization happens when a variable is given a value when it is created*
- *assignment replace an object's current value with a new one*

Definition 2.6 (list initialization)**.**

```
// initialization form
// = for copy initialize
// omit = for direct initialize

type var = value;
type var = {value}; // list initialization
type var{value};    // narrowing conversion, i.e. error convert double to int
type var(value);    // parenthesis initialization
```

Definition 2.7 (definition and declaration)**.**

```
// definition: allocate storage and define only once
// declaration: based type followed by a list of declarators specifying type and variable name

// headers should have guards to be safe to be included multiple times
#ifndef HEADER_H
#define HEADER_H
extern type var;
#endif
// source
#include "header.h"
type var = ...;
// main
#include "header.h"
```

2.2 Compound Types

Definition 2.8 (compound type)**.** *Compound type is defined in terms of another type.*

Definition 2.9 (reference)**.** *A reference defines an alternative name for an object.*

```
// reference must be initialized with object
type var = ...;
type &ref = var; // bind reference to its initializer
// reference cannot be rebind
```

Definition 2.10 (pointer)**.** *A pointer is an object that holds the address of another object.*

```
int i = 0;
int *p = &i; // p holds the address of i
*p           // return object i lvalue
p = nullptr; // null pointer: not bound to any object
p = 0;

// pointer to pointer
int **pp = &p; // * indicate pointer level

// ref to pointer
int *&ref = p; // refer to int*
```

2.3 Qualifier

Definition 2.11 (const)**.** *The compiler will replace the const var with its value.*

```
// const must be initialized
const int i = 1;

// const var is local to file by default
// must specify extern to be used in other files
// source.cpp
extern const i = 10;
// header.h
extern const i;
// main.cpp
#include "header.h"
i ...

// ref to const
// cannot change the object to which it bounds
const int i = 1;
const int &r = i;
```

```
const int &r2 = 2; // compiler: const int temp = 2; const int &r2 = temp;

// pointer to const
// cannot change the object to which it points
const int i = 1;
const int *p = &i;

// const pointer
// the address it holds cannot be changed
int i = 1;
int *const p = &i; // p always points to i
```

Definition 2.12 (constant expr)**.** *A constant expression is an expr whose value cannot change at compile time.*

```
constexpr int i = size();   // ok only if size is a constexpr function
constexpr int *p = nullptr; // const pointer, constexpr applies to p
```

Definition 2.13 (type alias)**.** *A synonym for a type.*

```
typedef int base;
using base = int;
typedef base *p;  // int*: pointer to int
p p1 = &i;

// const p: const modify the given type: int *const: constant pointer to int
const p p2 = 0; // int *const p2 = 0
const p *p3;    // int *const *p3
```

Definition 2.14 (const level)**.**

```
// top-level const: an object itself is const, i.e. const pointer
const int i = 1;
int *const p1 = &i;

// low-level const: point or refer to const object
const int *p2 = &i;
const int &r = i;

// when copying, top-level const is ignored,
// low-level const is never ignored
const int *const p3 = &i;
p2 = p3;
```

Definition 2.15 (auto)**.** *Deduce type from initializer at compile time.*

```
auto i = 1, *p = &i;   // must have initializer
auto i = 0, pi = 3.14; // error, types must be consistent

// auto may different from initializer's type
// auto ignores top-level const
```

Definition 2.16 (decltype)**.** *Return the operand type without evaluating the expr.*

```
decltype(expr) var
```

3 string

Definition 3.1 (string)**.** *A string is a variable-length sequence of characters.*

```
#include <string>
// using std::string;

// class defines how objects can be initialized
std::string s = "A";   // not include null char at the end of string literal
std::string s(N, 'c'); // N copies of char

// read string
std::cin >> s1 >> s2;

// read line
std::getline(std::cin, s);

// compare
// at least one should be std::string
std::string s = "Hello";
s < "Hiya" // compare first diff char ASCII: i > e
s < "Hello World"

// concat
// cant add two string literals, i.e. "A" + "B"
s + " World\n" // "Hello World\n"

// for range statement: iterate through elems in a given sequence
for (char c : s)
    ...
for (char &r : s) // change char in string
    r = toupper(r); // uppercase

// ispunct(c): check if char is punctuation, i.e. ',', '!', ...
// isspace(c): check if char is space

// subscript operator: s[n]
// !s.empty()
// s.size(): unsigned std::string::size_type
// s.size() < -1 => 2^32 - 1 for 32-bit int
// n: unsigned std::string::size_type
// 0 <= n < s.size()
for(decltype(s.size()) i = 0; i != s.size(); i = i + 1)
    ...

// iterator
```

4 vector

Definition 4.1 (vector)**.** *Variable-length sequence of objects of a given type.*

```
#include <vector> // class template
// using std::vector;

std::vector<int> v; // empty vector
std::vector<int> v = {val_1, ..., val_n};
std::vector<int> v(size, val); // {val, ..., val}
v.push_back(elem) // append elem

// for range statement
for(auto &r : v)
    ...
```

```
// compare

// subscript operator
// !v.empty()
// v.size(): unsigned std::vector<int>::size_type
// 0, ..., v.size() - 1
```

4.1 Iterators

Definition 4.2 (iterator)**.** *Access the elem in container. General mechanism, all lib containers have iterator.*

```
auto b = v.begin(); // return iterator: denotes first elem
auto e = v.end();   // return iterator: denotes one after the last elem

// deref
*b // return ref to elem obj
b->mem // arrow operator: (*b).mem, i.e. b->empty()

// operation
++b // next elem
--e // previous elem
b != e

// for vector and string iterators
auto mid = b + (e - b) / 2;
b <= e
```

5 Array

Definition 5.1 (array)**.** *Array is a fixed-size container of unnamed objects of a single type.*

```
// arrays are compound type
type a[N];
// cannot use auto to deduce type
// N is constexpr unsigned size_t
```

5.1 Array Declarations

Definition 5.2 (pointer, ref, and array)**.**

```
// cannot initialize or assign array with another array

// 1. array of pointers
type *a[N]; // array of N pointers to type

// 2. pointers or reference to array
type (*p)[N] = &a; // pointer to array of N type elements
type (&r)[N] = a;  // reference to array of N type elements

// 3. pointers or reference to array element
type *p = &a[0];
type *p = a; // a is type* pointer to first element

// auto return type is type* pointer to first element
auto p = a; // p = &a[0]
// decltype return type is array of N type elements
decltype(a) a2;
```

5.2 Access Elements of Array

Definition 5.3 (array element)**.**

```
// range for statement
for (auto &r : a)
    ...

// iterator
int *b = std::begin(a); // pointer to first elem
int *e = std::end(a);   // pointer to one after last elem

// subscript
// should define var to have unsigned size_t type
int *p = &a[m]; // p = a + m
p[n] // *(p + n) => a[m + n]

// C-style char string
char c[] = "str"; // c[] = {'s', 't', 'r', '\0'}

// initialize vector with array {a[i], ..., a[j]}
std::vector<int> v(a + i, a + j);
```

5.3 Arrays of Arrays

Definition 5.4 (multidimensional array)**.**

```
constexpr size_t rows=..., cols=...;
int a[rows][cols] = {0};

// subscript operator
a[n] // nth inner-array elem

// for range statement
// control vars except innermost array must be ref
for (auto &row : a)     // for every elem in outer array
    for(auto col : row) // for every elem in inner array
        ...

// pointers
int (*p)[cols] = a; // p = &a[0]

for (auto p = a; p != a + rows; ++p)       // int (*p)[cols] = &a[0]
    for (auto q = *p; q != *p + cols; ++q) // int *q = &a[0][0]
        ...

// type alias
using int_array = int[rows];
// typedef int int_array[rows];
for (int_array *p = a; p != a + rows; ++p)
    ...
```

6 Exprs

Definition 6.1 (expression)**.** *Composed of operators and operands.*

```
// unary operator: act one one operand
// & address-of
// * dereference
```

```
// binary operator: act on two operands1
// == equality
// * multiplication

// overloaded operator
// define operator in class

// lvalue and rvalue
// lvalue: can be assignment
// rvalue: cannot be assignment
// lvalue = ...                    return left-hand operand as lvalue
// address-of &lvalue              return a pointer to lvalue as rvalue
// deref * and subscript operator return lvalue
// ++iterator or --iterator       return lvalue
// decltype(lvalue)                return ref
// decltype(rvalue)                return the exact type of rvalue
decltype(*p) // int&
decltype(&i) // int*
decltype(&p) // int**

// compound expr with two or more operators
// precedence:
// operands of operators with higher precedence group tightly than
// operands of operators at lower precedence
// associativity: group operands with same precedence
// left associative: operators with same precedence group from left-most to right
// right associative: operators with same precedence group from right-most to left
// () force grouping

// arithmetic operator
// left associative
// return rvalue
// reminder: int % int
// return same sign as m
(-m) % n // -(m % n)
m % (-n) // -(m % n)

// logical and relational operator
// return bool val
// 0 return false
// non-zero return true
// guarantee eval order
expr1 && expr2 // eval expr2 if and only if expr1 is true
expr1 || expr2 // eval expr2 if and only if expr1 is false
// precedence:
// right associative
// 1. !
// left associative
// 2. <=, >=
// 3. ==, !=
// 4. &&
// 5. ||
i != j < k // i != (j < k)

// assignment operator
// low precedence
// right associative
```

```
// left-hand operand: a modifiable lvalue
// return left-hand operand lvalue
// compound assignment operator
// *=, /=, %=
a *= b;    // compound assignment: a eval only once
a = a * b; // ordinary assignment: a eval twice

// increment and decrement operator
// require lvalue operand
// prefix:  increment operand and return the changed object as lvalue
// postfix: increment operand and return object's original value as rvalue
// use postfix only when necessary
// precedence: ++-- > dereference
*b++ // *(b++)

// evaluation order

// member access operator
p->mem => (*p).mem
// precedence: . > dereference
pointer-> return lvalue
.mem return lvalue or rvalue same as mem

// conditional operator
cond ? expr1 : expr2;
// return lvalue if both expr1 and expr2 are lvalue
// otherwise return rvalue
cond ? expr1 : cond ? expr2 : expr3
// right associative
// low precedence, need ()

// bitwise operator
// hardware

// shift operator (IO operator)
// left associative
// precedence:
1. arithmetic operator
2. IO operator
3. relational, assignment, conditional operator

// sizeof operator
// right associative
// return constexpr size_t in byte
// 1 byte = 8 bits
sizeof(type)
sizeof(expr)
// array
sizeof(a) // sizeof(*a) * size

// comma operator
// guarantee eval order from left to right
// lowest precedence
// expr1, expr2
// return expr2 lvalue or rvalue
for (int v1=..., v2=...; cond; expr1, expr2)
    ...
```

Definition 6.2 (IO objects).

```
#include <iostream>
// using std::cout, std::cin;
std::cout << "str"; // return left operand: std::cout
std::cin  >> v;     // return left operand: std::cin
// IO operators: left associative
std::cout << "str" << "\n"; // (std::cout << "str") << "\n";
std::cin  >> v1 >> v2;      // (std::cin >> v1) >> v2;

while(std::cin >> v)
    ...
// istream become invalid when
// 1. reading a non-integer val: 1 2 3.1
// 2. end-of-file: Ctrl-z RET (Windows), Ctrl-d RET (Linux)
```

6.1 Type conversion

Definition 6.3 (type conversion)**.**

```
// implicit conversion
// arithmetic conversion
// integer promotion
// unsigned type
// array to pointer conversion
// conversion to bool
// conversion to const

// explicit conversion
// static_cast<type>(expr)
int i = 1, j = 2;
// double k = i / j; // 0
double k = static_cast<double>(i) / j; // 0.5
// const_cast: remove low-level const
// reinterpret_cast: bit-pattern of operands
// old-style cast
// type (expr);
// (type) expr;
```

7 Statement

Definition 7.1 (statement)**.**

```
// simple statement
// expr statement
; // null statement
// compound statement (block, scope)
// {} empty block is equivalent to ; null statement

// flow-of-control statement
// conditional statements
// if statement
if (cond)
    statement;
else if (cond)
    statement;
else
    statement;

// switch statement
switch (v)
```

```
{
    case expr1: // constexpr
        statement;
        break;
    case expr2:
        statement;
        break;
    ...
    default:   // no case match
        statement;
        break;
}

// iterative statement
while(cond)
    statement;

// traditional for statement
for (init-statement; cond; expr) // loop var used locally inside for body
    statement;

// range for statement
// iterate through elems of a container or sequence
for (declaration : sequence) // ref to write elem
    statement;

// do while statement
do
    statement
while (cond);

// jump statements:
// break statement
break; // terminate nearest loop or switch

// continue statement
// terminate current iteration of nearest loop and begin next iteration
continue;

// goto statement
label:
    ...
goto label;

// statement scope
// vars defined in control structure are only visible within scope
```

7.1 Try and Exception

Definition 7.2 (exception)**.**

```
// try block
try
{
    throw std::runtime_error("ERROR MESSAGE");
}
catch (std::runtime_error e)
{
    std::cout << e.what() << std::endl; // ERROR MESSAGE
```

```
}

// exceptions
```

8 Function

Definition 8.1 (func). *Function is a block of code with name.*

```
type func(params) // single declarator: type v1, type v2, ...
{
    return value;
}

// call operator
// 1. initialized func params from arguments
// 2. execute calling function -> execute called function ->
// ends when encountering return statement ->
// return value -> back to calling function ->
// initialize call expression with returned value
func(arguments);
```

Definition 8.2 (lifetime).

```
// lifetime for local variables
// automatic object
// params destroyed after function call

// local static object
// not destroyed until program terminates
static type var = ...;
```

8.1 Func Declaration

Definition 8.3 (func prototype).

```
// func must be declared before we can use

// header
type func(params); // semicolon replaces func body in decl

// source
// should include header so that compiler will verify func def and decl are consistent
#include "func.h"
type func(params) {...}

// separate compilation
g++ -c main.cpp   // generate object file main.o
g++ -c func.cpp   // func.o
g++ main.o func.o // link object files
```

8.2 Parameter Initialization

Definition 8.4 (param initialization).

```
// C++ use reference param to avoid copies
type& param = argument // param bound to argument

// nonreference
type param = argument  // argument value is copied
// C use pointer param to access object
```

```
type* param = &argument

// const param
// top-level const are ignored
type param = const argument
// low-level const cannot be ignored
const type& param = const argument
const type* param = &const argument
```

Definition 8.5 (array param)**.**

```
// array param
// cannot copy array
type array[N] = {};
type func(const type* param);   // array converted to &array[0]
type func(const type param[]);  // equivalent to const type* param
type func(const type param[M]); // compiler ignores first dimension
// cannot use type var[] = array in expr!

// when func doesn't need write access to array elements,
// array param should be pointer to const

// array size rely on additional inform:
// C-style string stops at null char
type func(const char* cp) {if(cp) ...}
// beg and end pointer to first and last element in array
const int* beg = std::begin(array)
const int* end = std::end(array)
// define a second param indicates array size
type func(const int[], size_t);

// array ref param
int (&param)[N] = array // type size limits usefulness of func

// multidimensional array param
// the second and subsequent dimensions must be specified
type func(int (*matrix)[Y][Z][...], int X);
type func(int matrix[][Y][Z][...], int X); // compiler ignores first dimension

// varying number of params
// std::initializer_list: template type
void func(params, std::initializer_list<T> il);
// number of arguments of a single type in il
// elems in il are always const
void error_msg(std::initializer_list<std::string> il)
{
        for (auto b = il.begin(); b != il.end(); ++b)
        std::cout << *b << " ";
    }

error_msg({"FUNC", s1, s2});

// ellipsis params
void func(params, ...)
```

8.3 Return Statement

Definition 8.6 (return)**.** *Terminate func and return control to func call.*

```
// return statement
```

```
// analogous to break statement
return;       // return with no value used in void func(params);
return value; // type tmp = value
// return value is used to initialize a temporary as result of func call

// return reference
// funcs that return reference are lvalues
// other return types are rvalues

// call operator
// same precedence as dot and arrow operator
// call operator is left associative
func(argument).member_func() // return class type object

// return list
std::vector<type> tmp = {...}

// cannot return array or func
// return pointer or reference to array or func
type (*func(params)) [N]; // return pointer to array of N type elements

// use typedef
typedef int arrayT[N];    // synonym to array of n ints
using arrayT = int[N];
arrayT* func(params);
// use decltype
int a[] = {1,2,3};
decltype(a) *func(params);
// use trailing return type
auto func(params) -> int(*)[N];

// main return
// 0 indicates success
// nonzero has machine-dependent meaning
// machine-independent preprocessor variable
// EXIT_SUCCESS
```

8.4 Recursion

Definition 8.7 (recursion)**.** *A function that calls itself is a recursive function.*

8.5 Pointers to Funcs

Definition 8.8 (function pointer)**.** *Function pointer points to func type determined by return and params types.*

```
type func(params);
type (*p)(params);

p = func;        // func is automatically converted to pointer
// p = &func;    // equivalent assignment
p(params);       //call func
// (*p)(params); // equivalent call
p = nullptr;     // point to no func
// p = 0;

// func pointer as param
type func(params, type p(params));
// type func(params, type (*p)(params)); // equivalent decl
```

```
// typedef
typedef type F(params);      // func type
using F = type(params);      // equivalent func type
typedef decltype(func) F;    // equivalent func type

typedef type (*PF)(params);  // pointer to func type
using PF = type(*) (params); // equivalent pointer to func type
typedef decltype(func) *PF;  // equivalent pointer to func type

PF p;
F *p;

// def func return pointer to func
PF func(params);
F *func(parmas);
type (*func(params)) (params); // directly decl
auto func(params) -> type (*) (params);
decltype(func2) *func(params);
```

8.6 Overloaded Funcs

Definition 8.9 (overload)**.** *Same name funcs with different params list in one scope.*

```
// error for two funcs to differ only in terms of return types

// top-level const is indistinguishable from one without a top-level const
// except low-level const
// const_cast

// func matching
// 1. candidate funcs: all funcs
// 2. viable funcs: arguments match params
// 3. best match: exact match is better than argument type conversion
// argument type conversion:

// use small func inside overloaded funcs to avoid writing same code in more than one place
```

8.7 Default Arguments

Definition 8.10 (default argument)**.** *Func with default argument can be called with without that argument.*

```
// func.h
// default arguments ordinarily should be specified with func declaration in header
type func(params, def_defaults_for_params);
```

8.8 inline and constexpr Func

Definition 8.11 (inline func)**.** *Calling a small func is slower than equivalent expr. A inline function is expanded in line at each call to avoid func call overhead.*

```
inline type func(params);
// inline is only a request to compiler.
// The compiler may choose to this request.

// inline and constexpr func defined in header
```

Definition 8.12 (constexpr func)**.** *Can be used in constant expression.*

```
constexpr type func(params);
// implicitly inline
// replace a call with return value
// return type must be literal type
// func body contains exactly one return statement
```

8.9 Command-Line Options to main

Definition 8.13 (argument to main)**.** *Use argument to main to guid operation of program.*

```
int main(int argc, char *argv[]) {...}
int main(int argc, char **argv) {...}

// argc: size of argv
// argv[0]: program name
// argv[1]: option1
// argv[2]: option2
// ...
```

8.10 Debugging

Definition 8.14 (assert and NDEBUG)**.** *Debugging code is executed only during developing.*

```
// #define NDEBUG // turn off assert, defined before #include <cassert>
#include <cassert>
assert(expr); // eval expr. if expr is false, write a message and terminate program.

// preprocessor macro
// preprocessor names are managed by preprocessor not compiler
// use preprocessor name directly and do not provide using decl

// conditional debugging using NDEBUG
void func(int i)
{
#ifndef NDEBUG
    std::cerr << __func__ << ": i is " << i << std::endl;
#endif
}
// local static defined by compiler
// __func__: func name
// __FILE__: file name
// __LINE__: current line number
// __TIME__: compiled time
// __DATE__: compiled date
```

9 class

https://www.learncpp.com/

Definition 9.1 (OOP)**.**

Definition 9.2 (class)**.** *Data abstraction and encapsulation: separation of interface and implementation*

- *interface: operations that users can execute*
- *implementation: class data members and funcs*

```
// header
class MyClass
{
```

```
public:
    friend type nonmember_func(params);
    // allow nonmember func to access class nonpublic members
    // friends are not affected by access control
    // group friend decls at beginning or end of class
    // friend class MyClass2; // access all member funcs in MyClass2
    // friend type MyClass2::member_func(params);

    // member funcs access object through implicit parameter this
    // any use of member of class inside member func is an implicit reference through this
    type func(params) const; // const member func indicates that this is a pointer to const
    // const member func cannot change data members of object on which it is called

private:
    // class cannot have data members of its own type
    // class can have data members that are pointers or references to its own type
    type var;
}; // must end with ;

// declare friend in class header
// friend declaration inside class only affects access
// even if friend func can be defined inside class,
// func declaration must still be provided outside class
type nonmember_func(params);

// source
type MyClass::func(params) const {...}
type nonmember_func(params) {...}

// main
MyClass myObj;
```

9.1 Additional Class Features

Definition 9.3 (this)**.**

```
// funcs return reference are lvalue objects
type func(params) { return *this; }

const type func(params) const { return *this; }
// const member func return *this as reference should return a reference to const

// member func can be overloaded based on whether it is const
```

Definition 9.4 (type member)**.**

```
class MyClass
{
    typedef type T; // type member must appear before they are used
    // appear at beginning of class

    T var;
}
```

Definition 9.5 (inline member func)**.**

```
// member funcs defined inside class are automatically inline
// member funcs defined outside class need to be specified inline
class MyClass
{
```

```
    type func(params);
}

inline type func(params) {...}
// inline member func should be defined in class header
```

Definition 9.6 (mutable data member)**.** *A mutable data member is never const.*

```
// const member func can change a mutable member
mutable type var;
```

9.2 Constructor

Definition 9.7 (constructor)**.** *Class initialization, copy, move, assignment, destruction.*

```
class MyClass
{
public:
    // constructor funcs
    // initialize data members
    // same name as class
    // no return type

    // default constructor
    // direct initialization
    // automatically called when an obj is created
    MyClass() {}
    MyClass(params) member(expr), ... {...}

    // copy constructor
    // copy initialization
    // copy each nonstatic mem from given object into one being created
    // first param is ref to class
    MyClass(const MyClass &);

private:
    // in-class initializer
    // = for copy initialization
    // braces for direct initialization
}
```

9.3 Operator

Definition 9.8 (operator)**.**

9.4 Inheritance

Definition 9.9 (inheritance)**.**

9.5 Template

Definition 9.10 (template)**.**

10 Dynamic Memory

11 Others

Definition 11.1 (comment)**.**

```
// single line
/* pair cannot nest */

std::cout << /* "*/" /* "/*" */; // std::cout << " /* ";
// 1. from right to left
// 2. find */, remove everything between */ and left /*
// 3. repeat step 1-2
```

References

[1] S.B. Lippman, J. Lajoie, and B.E. Moo. *C++ Primer*. Addison-Wesley Professional, 2012. ISBN: 978-0-321-71411-4.

[2] Shapira Yair. *Solving PDEs in C++*. 2nd Edition. Philadelphia, PA: Society for Industrial and Applied Mathematics, 2012. URL: https://epubs.siam.org/doi/abs/10.1137/1.9781611972177.